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Primary Drivers of FDA Complete Response Letters: Insights from a SAS® - Based Analysis

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ABSTRACT

Complete Response Letters (CRLs) document deficiencies identified when the U.S. Food and Drug Administration (FDA) cannot approve a New Drug Application (NDA) or Biologics License Application (BLA) in its current form. The narrative format complicates systematic analysis. This project used SAS® software to structure and analyze 401 CRLs dated 2002–2026 from FDA's approved and unapproved CRL folders. These letters represent 298 FDA application numbers. Manual coding identified 915 individual deficiencies; SAS transformations produced 610 unique CRL-domain records and letter-level domain indicators. Of the 610 unique CRL-domain records, 314 (51.5%) were CMC and 147 (24.1%) were Clinical. CMC was present in 314 of 401 letters (78.3%), and 160 letters (39.9%) contained more than one deficiency domain. Co-occurrence summaries characterized both cross-domain and CMC-subtype combinations. Exploratory logistic models found no statistically significant application-type association with CMC presence, while application type and CRL year were associated with multi-domain findings. Clustered generalized estimating equation (GEE) sensitivity analyses accounting for repeated CRLs associated with the same application number produced the same point estimates with wider confidence intervals, and the multi-domain associations remained significant. The workflow produced datasets at the deficiency, CRL-domain, and letter levels, along with summaries and dashboard exports.

INTRODUCTION

FDA Complete Response Letters communicate deficiencies that must be addressed before a New Drug Application (NDA) or Biologics License Application (BLA) can be approved. Under 21 CFR § 314.110 (NDAs) and § 601.3 (BLAs), a complete response letter describes the deficiencies FDA has identified when an application cannot be approved in its present form. Letters may cite multiple issues spanning Chemistry, Manufacturing, and Controls (CMC), Clinical, Regulatory, Nonclinical, Statistics, and Patent Certifications. Although the letters are informative individually, their narrative structure makes it difficult to compare patterns across documents and application types. Sacks et al. (2014) used FDA review materials to characterize the reasons novel drugs reviewed by FDA's drug center were delayed or denied approval between 2000 and 2012; this project uses a different set of CRLs that also includes BLAs.

The purpose of this project was to build a reproducible SAS workflow for manually coded CRL data and to describe deficiency prevalence, subtype patterns, and co-occurrence within letters. In the title, "drivers" refers to the deficiencies documented in these letters, not causes of CRL issuance.

DATA MANAGEMENT AND WORKFLOW

Data preparation

The CRL collection was drawn from FDA's approved (288 letters) and unapproved (113 letters) CRL folders, released through FDA's transparency efforts and accessed in 2026; the folder was retained as source metadata. FDA notes that additional archival batches may be added over time. As such, the 401 letters used in this analysis are a curated collection from a specified time frame, not every CRL ever issued by the FDA. The project workbook contains a CRL_Events sheet with one row per letter and a CRL_DEF sheet with one row per coded deficiency. Each CRL received a distinct event_id linking its deficiency rows, while application_number identifies the underlying FDA application, which may have multiple CRLs (up to six here). Recorded variables include application type, product, sponsor, review office, regulatory pathway, and CRL year. The recorded years span 2002–2026; 2026 represents a partial year at the time of data assembly. In total, the dataset contains 401 letters associated with 298 application numbers and 915 deficiency rows (U.S. Food and Drug Administration, 2025).

All 915 deficiency rows were manually coded using predefined domain and subtype categories. Where a CRL clearly identified a deficiency through section headers or explicit labeling, the corresponding category was assigned directly. Any ambiguous cases were reviewed with input from a pharmaceutical industry professional with senior experience in CMC regulatory affairs. When a domain was identifiable but subtype detail was redacted or insufficiently specific, the

subtype was labeled “Redacted”; 76 deficiency rows received this label. Office and regulatory-pathway information were not available for every letter, so analyses using those variables were treated descriptively.

Table 1. Units of analysis used in the SAS workflow.

Dataset / unit	Records	Purpose
CRL_Header / letter	401	One row per letter; application-level characteristics
CRL_Deficiency / finding	915	One row per coded deficiency and subtype
Domain_Unique / CRL-domain	610	One row per event_id × domain, avoiding repeated domain counts
Domain_Event / letter	401	One row per letter with six binary domain indicators

SAS workflow

Microsoft Excel was imported through the XLSX LIBNAME engine to create permanent SAS header and deficiency datasets. PROC SORT and the DATA step linked these data by event_id into CRL_Long. For domain-level analyses, PROC SORT with NODUPKEY retained each event_id–deficiency_domain combination once; this avoided counting several CMC subtypes within the same letter as multiple CMC-domain occurrences. SAS DATA steps then produced letter-level domain flags and separate CMC subtype flags. PROC FREQ summarized frequencies and co-occurrence, PROC LOGISTIC modeled selected binary letter-level outcomes, and PROC GENMOD provided clustered GEE sensitivity analyses for repeated CRLs within application number. ODS PDF and PROC EXPORT produced report and CSV outputs, with CSVs used by the R Shiny dashboard.

Example 1. SAS code used to retain one record per letter and deficiency domain.

```
PROC SORT DATA=sasdata.crl_long
    (KEEP=event_id deficiency_domain application_type
          regulatory_pathway year office)
    OUT=sasdata.domain_unique NODUPKEY;
    BY event_id deficiency_domain;
RUN;
```

Example 2. DATA step used to define multi-domain status from the six letter-level domain flags.

```
data domain_event_model2;
    SET sasdata.domain_event_model;
    domain_count = sum(cmc_f, clinical_f, regulatory_f,
                       nonclinical_f, stats_f, patent_f);
    IF domain_count > 1 then multi_f = 1;
    ELSE multi_f = 0;
RUN;
```

Every percentage below is based on one of three counts: deficiency rows (n = 915), unique letter-domain records (n = 610), or letters (n = 401).

ANALYTICAL METHODS

PROC FREQ summarized coded domains and subtypes. The domain-by-application-type table compares the distribution of 610 unique letter-domain records descriptively; no conventional Pearson chi-square p-value is reported for this table because letters can contribute more than one domain. Letter-level presence indicators were created for each major domain, and multi-domain letters were defined as those containing two or more distinct domains. Exact combination summaries describe the complete domain pattern within each letter, while pairwise tables identify letters containing both members of a selected domain or subtype pair, even if other deficiencies were also present.

Three exploratory PROC LOGISTIC models used CRL letters as observations: an unadjusted model of CMC presence by application type, a CMC model adjusted for CRL year, and a model of multi-domain status by application type and CRL year. NDA served as the reference application type. Regulatory pathway was excluded from the models because, in this dataset, 351 pathways corresponded to BLAs and 505 pathways to NDAs, creating substantial overlap with application type; pathway information was also unavailable for approximately one quarter of CRL-domain records. Odds ratios (ORs), Wald 95% confidence intervals (CIs), and p-values are reported. Because 72 of 298 application numbers contributed more than one CRL, three models were rerun as sensitivity analyses using PROC GENMOD with generalized estimating equations, an independent working correlation structure, and REPEATED SUBJECT=application_number. Empirical (robust) standard errors from these clustered models were used to assess whether within-application dependence materially changed inference.

Example 3. Clustered GEE sensitivity model for multi-domain CRLs.

```
PROC GENMOD DATA=domain_event_model2 DESCENDING;
    CLASS application_number application_type(REF='NDA') / PARAM=REF;
    MODEL multi_f = application_type year / DIST=BINOMIAL LINK=LOGIT;
    REPEATED SUBJECT=application_number / TYPE=IND;
    ESTIMATE 'BLA vs NDA' application_type 1 / EXP;
RUN;
```

RESULTS

Domain frequencies and subtypes

CMC accounted for 314 of 610 unique CRL-domain records (51.5%), followed by Clinical (147; 24.1%), Regulatory (64; 10.5%), Nonclinical (61; 10.0%), Statistics (13; 2.1%), and Patent Certifications (11; 1.8%). At the letter level, 314 of 401 CRLs (78.3%) contained at least one CMC deficiency. These percentages answer different questions: domain-record percentages sum to 100%, while letter-level presence percentages can overlap because a CRL can contain several domains.

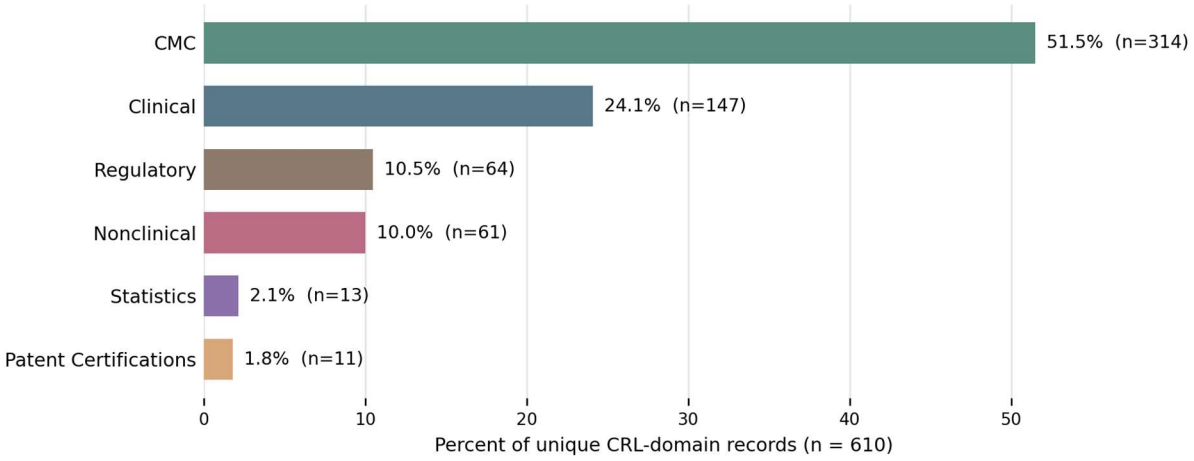


Figure 1. Distribution of unique CRL-domain records by deficiency domain (n = 610); repeated findings in the same domain within one letter count once.

Within the 549 individual CMC deficiency rows, Facility was the most frequently coded subtype (232; 42.3%), followed by Specifications (81; 14.8%) and Manufacturing Controls (79; 14.4%). The leading Clinical subtype was Efficacy (63 of 203 Clinical deficiency rows; 31.0%); the leading Regulatory subtype was Labeling (47 of 69; 68.1%). In Nonclinical findings, Insufficient Nonclinical Support and Toxicology had the same frequency (20 of 68 each; 29.4%). Subtype percentages are calculated from deficiency rows within each parent domain, not from all 401 CRLs.

NDA versus BLA descriptive patterns

There were 306 NDA letters and 95 BLA letters. The 610 unique letter-domain records comprised 493 NDA and 117 BLA records. CMC contributed a larger share of BLA domain records (81/117; 69.2%) than NDA domain records (233/493; 47.3%). At the letter level, CMC appeared in 81 of 95 BLA letters (85.3%) and 233 of 306 NDA letters (76.1%).

Table 2. Distribution of unique CRL-domain records by application type (n = 610). Percentages are calculated within application type.

Domain	NDA, n (%)	BLA, n (%)
CMC	233 (47.3)	81 (69.2)
Clinical	125 (25.4)	22 (18.8)
Regulatory	54 (11.0)	10 (8.5)
Nonclinical	60 (12.2)	1 (0.9)
Statistics	10 (2.0)	3 (2.6)
Patent Certifications	11 (2.2)	0 (0.0)
Total	493 (100)	117 (100)

Domain co-occurrence

Among 401 letters, 160 (39.9%) contained two or more distinct deficiency domains. The most frequent exact patterns were CMC only (172; 42.9%), CMC plus Clinical only (51; 12.7%), and Clinical only (49; 12.2%). The CMC-plus-Regulatory-only pattern appeared in 27 letters (6.7%). In the pairwise table, 80 letters contained both CMC and Clinical, including letters that also contained additional domains; this differs from the 51 letters that cited only those two domains.

Table 3. Common exact domain combinations among 401 CRLs.

Exact pattern	CRLs	Percent of letters
CMC only	172	42.9%
CMC + Clinical only	51	12.7%
Clinical only	49	12.2%
CMC + Regulatory only	27	6.7%
CMC + Nonclinical only	21	5.2%
CMC + Clinical + Nonclinical	11	2.7%

CMC subtype co-occurrence

Among the 314 letters containing CMC, pairwise subtype analysis found 45 with both Facility and Manufacturing Controls, 40 with Facility and Specifications, and 35 with Specifications and Manufacturing Controls. These are inclusive pairwise counts: a letter with three or more CMC subtypes may contribute to multiple pairs. The most common exact pattern was Facility alone, in 135 of 314 CMC letters (43.0%). Overlap was strongest between Specifications and Manufacturing Controls (35 of the 75 Specifications letters, 47%) and between Manufacturing Controls and Stability (33 of 74, 45%); by comparison, 45 of 226 Facility letters (20%) also had Manufacturing Controls. Redacted subtype labels were kept as their own category and not interpreted as a deficiency type.

Exploratory logistic regression

The unadjusted CRL-level CMC model estimated higher odds of CMC presence in BLA than NDA letters (OR = 1.813, 95% CI 0.970–3.387), but the association did not meet the 0.05 threshold (p = 0.0622). After adjustment for CRL year, application type (OR = 1.780, p = 0.0815) and year (OR = 1.007, p = 0.8369) were not statistically significant predictors of CMC presence. Clustered GEE sensitivity models yielded the same point estimates but wider empirical CIs: the unadjusted BLA association was OR = 1.813 (95% CI 0.895–3.673; p = 0.0988), and the adjusted BLA association was OR = 1.780 (95% CI 0.869–3.649; p = 0.1152). Year remained non-significant in the clustered CMC model (OR = 1.007, 95% CI 0.924–1.096; p = 0.8780).

In the exploratory multi-domain model, BLA letters had lower modeled odds of containing multiple domains than NDA letters after adjusting for year (OR = 0.375, 95% CI 0.214–0.657; p = 0.0006). Later CRL year was also associated with lower modeled odds of multi-domain findings (OR per year = 0.937, 95% CI 0.886–0.991; p = 0.0218). After clustering by application number, both associations remained statistically significant: BLA versus NDA OR = 0.375 (95% CI 0.206–0.684; p = 0.0014) and year OR = 0.937 (95% CI 0.879–0.998; p = 0.0443). The year association was attenuated after accounting for repeated CRLs. On average, BLA letters cited 1.2 domains and NDA letters 1.6 (117/95 and 493/306).

Table 4. Primary PROC LOGISTIC results for CMC presence and multi-domain status (401 CRL letters).

Outcome / model	Predictor	OR (95% CI)	p-value
CMC, unadjusted	BLA vs NDA	1.813 (0.970–3.387)	0.0622
CMC, adjusted	BLA vs NDA	1.780 (0.930–3.406)	0.0815
CMC, adjusted	CRL year	1.007 (0.945–1.073)	0.8369
Multi-domain, adjusted	BLA vs NDA	0.375 (0.214–0.657)	0.0006
Multi-domain, adjusted	CRL year	0.937 (0.886–0.991)	0.0218

DASHBOARD REPORTING

CSV exports from the SAS workflow feed an R Shiny dashboard for exploring coded CRLs. The dashboard includes filters for application type, year, and deficiency domain; domain and subtype summaries; CMC-specific views; and downloadable tables. The dashboard interface is illustrated in Figure 2.

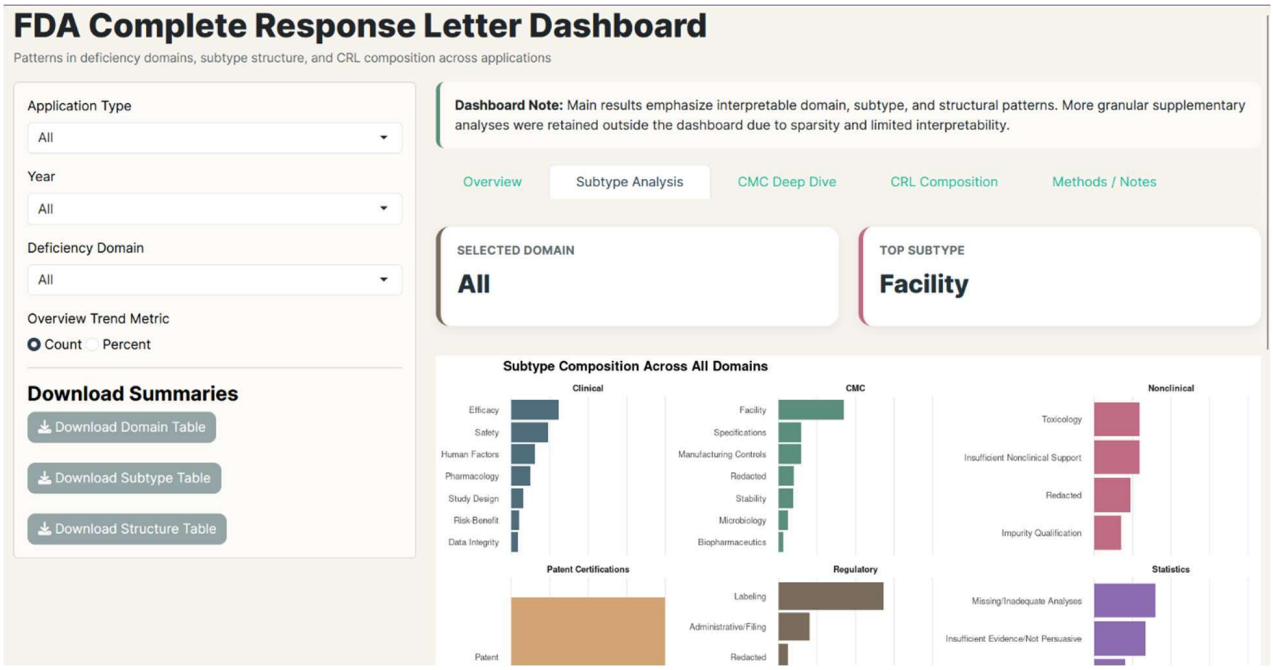


Figure 2. R Shiny dashboard interface for reviewing coded CRLs. Project URL: https://hannahweber-crl-deficiency-analysis.share.connect.posit.cloud/

DISCUSSION

CMC was the most frequent coded domain: 51.5% of unique letter-domain records, and 78.3% of letters contained CMC deficiencies at least once. Within CMC, Facility, Specifications, and Manufacturing Controls were the most frequent coded deficiency rows (392 of 549; 71.4%). These are counts within this CRL sample and should not be taken as evidence that CMC deficiencies are a more common cause of CRLs across all FDA applications.

CMC and Clinical appeared together in 80 letters when other domains were allowed, but in only 51 as the sole pair. Within CMC, Facility was frequently cited alone (135 of 226 Facility letters), while the strongest overlaps were Specifications with Manufacturing Controls (35 of 75 Specifications letters) and Manufacturing Controls with Stability (33 of 74 Manufacturing Controls letters).

The regressions found no significant application-type difference in CMC presence. The multi-domain associations with application type and year held up under clustered GEE models, although the year effect weakened from $p = 0.0218$ to $p = 0.0443$. The year result should be interpreted with caution, since the number of letters per year depends on which letters FDA has released. The main practical point of the workflow is keeping deficiency rows, CRL-domain records, and letters separate, because each gives a different percentage.

LIMITATIONS

This observational study covers a curated collection from FDA's approved and unapproved CRL folders, not all FDA submissions or all CRLs ever issued; record availability and inclusion may affect observed proportions and year patterns, and 2026 is only partially observed. Some fields are missing, and 76 individual deficiency rows have redacted subtype details. Coding was performed manually using predefined categories, with ambiguous cases reviewed with an experienced CMC regulatory professional; however, formal inter-rater reliability was not assessed. Seventy-two of 298 application numbers are associated with more than one CRL. Primary PROC LOGISTIC models treated letters as observations, while sensitivity analyses using GEE clustered by application number accounted for within-application dependence; conclusions were generally consistent across approaches. The models also discriminate poorly (c-statistics 0.53–0.63), so the associations are weak. No comparison group of applications without CRLs was included; consequently, the study cannot estimate causal drivers or the probability of receiving a CRL.

CONCLUSION

A SAS-based workflow transformed 401 curated CRL letters and 915 coded deficiencies into analysis-ready datasets at the deficiency, unique letter-domain, and letter levels. CMC was the most common coded domain, nearly 40% of letters contained multiple domains, and the co-occurrence analyses highlighted recurring domain and CMC-subtype combinations. Clustered GEE sensitivity analyses showed that the multi-domain associations with application type and year persisted after accounting for repeated CRLs within application number. Together, the SAS workflow, reporting, and dashboard outputs provide a foundation for future work on sequential CRLs, persistence of deficiencies, and regulatory outcomes once the necessary longitudinal and outcome data are validated.

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